Indian Institute of Technology, Kanpur
Proposal for a New Course

1. Course No: SPA***A

2. Course Title: Mathematical Methods in Space Sciences & Engineering

3. Lectures per week: 3 (L), Tutorial: 0 (T), Laboratory: 0 (P), Additional hours: (0-2): 0 (A), Credits (3*L+2*T+1*P+A): 5
   Duration of Course: Half Semester

4. Proposing Department: Space, Planetary and Astronomical Sciences and Engineering.

5. Proposing Instructor: Pankaj Jain

6. Other Instructors who may teach this course: Sharvari Nadkarni-Ghosh, Kartick Sarkar, Rohit Sharma, Ishan Sharma

7. Course Description (A) Objectives: The course aims to introduce students to numerical techniques used in Space Science & Astronomy.

   (B) Contents (preferably in the form of 5 to 10 broad titles):

   1. **Vector Analysis** (3 lectures)
      Vector differential calculus, gradient, divergence and curl, surface and volume integrals
   2. **Matrix Algebra** (6 lectures)
      Introduction to matrices, diagonalization, solution of linear equations using matrices.
   3. **Coordinate Systems** (5 lectures)
      Curvilinear coordinates, coordinate transformation, applications to astronomy
   4. **Ordinary differential equations** (6 lectures)
      First and second order ODEs and their solutions. Separation of variables method to solve PDEs and various types of boundary conditions - initial value problem vs boundary value problem etc.

   (C) Pre-requisites, if any: N/A

   (D) Short summary for including in the Courses of Study Booklet: vecor analysis, vector dIfferential calculus, linear vector spaces, matrices, tensors, coordinate systems, Astronomical coordinate system transformation, ordinary differential equations, examples from Astronomy

7. Recommended Books:
   - Mathematical Methods in Classical and Quantum Physics, T. Dass & S. K. Sharma
   - Mathematical Methods for Physicists, G. B. Arfken, H. J. Weber and F. E. Harris, Elsevier, seventh edition.
   - Advanced Engineering Mathematics, E. Kreyszig, John Wiley & Sons, tenth edition.
8. Any other remarks:

Dated: Proposer:

Dated: DUGC/DPGC Convener:

The course is approved/not approved

Chairman, SUGC/SPGC

Dated: